INCH-POUND
MIL-DTL-23053/14B
17 June 1996
SUPERSEDING
MIL-I-23053/14A
01 May 1986

DETAIL SPECIFICATION SHEET

INSULATION SLEEVING, ELECTRICAL, HEAT SHRINKABLE, ETHYLENE-TETRAFLUOROETHYLENE FLUOROPOLYMER, SEMI-RIGID

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the sleeving described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-DTL-23053.

REQUIREMENTS:

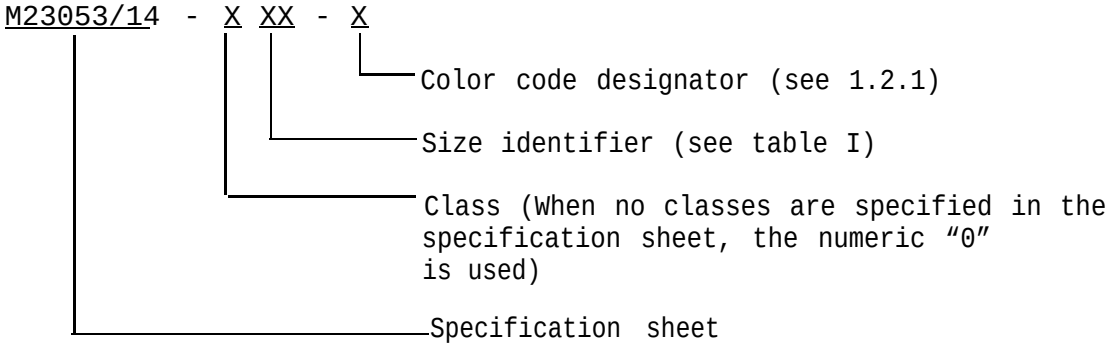
Polymer type: The base polymer used in formulating this sleeving shall be ethylene-tetrafluorethylene fluoropolymer.

Continuous operating temperature range: -100°C (-148°F) to 150°C (302°F).

Color: The sleeving shall be furnished in clear (translucent white to transparent blue), the as manufactured color.

Longitudinal change: ± 10 percent

Military part number: The Military part number shall consist of the basic number of this specification sheet and dash numbers as follows:



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Example: 0.500 inch (12.7 mm) as supplied ID ETFE sleeving shall be identified as M23053/14-006-C.

TABLE I. Construction details, inches (mm). 1/

Military part number	As supplied ID minimum	After unrestricted shrinkage	
		ID maximum	Wall thickness 2/
M23053/14-001-C	.093 (2.4)	.062 (1.6)	.010 ± .002(.25 ± .05)
M23053/14-002-C	.125 (3.2)	.083 (3.1)	.010 ± .002(.25 ± .05)
M23053/14-003-C	.187 (4.7)	.125 (3.2)	.011 ± .002(.28 ± .05)
M23053/14-004-C	.250 (6.4)	.166 (4.2)	.013 ± .003(.33 ± .08)
M23053/14-005-C	.375 (9.5)	.250 (6.4)	.013 ± .003(.33 ± .08)
M23053/14-006-C	.500 (12.7)	.345 (8.3)	.013 ± .003(.33 ± .08)
M23053/14-007-C	.750 (19.0)	.500 (12.7)	.018 ± .004(.46 ± .10)
M23053/14-008-C	1.000 (25.4)	.665 (16.7)	.022 ± .004(.56 ± .10)
M23053/14-009-C	1.250 (31.8)	.835 (21.2)	.030 ± .004(.80 ± .10)
M23053/14-010-C	1.500 (38.0)	1.000 (25.4)	.030 ± .004(.80 ± .10)

1/ Diameter limits for object to be enclosed shall be as recommended in technical data.

2/ Wall thickness dimensions are less when shrinkage is restricted.

Unrestricted shrinkage: Test method 4.6.5; 175° ± 1°C (347° ± 2°F) for 10 minutes.

TABLE II. Physical properties.

Characteristic	Requirement	Test procedure and conditions
As supplied:		
ID minimum	Table I	4.6.3
Restricted shrinkage	No cracks	4.6.6.12 175° ± 2°C (347° ± 4°F)
Voltage withstand	Pass	4.6.6.2
Clarity stability	Pass	4.6.16 175° ± 1°C (347° ± 2°F), 24 hours

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TABLE II. Physical properties. - Continued

Characteristic	Requirement	Test procedure and conditions
After unrestricted shrinkage:		
ID, maximum	Table I	4.6.3
Wall thickness	Table I	4.6.3
Tensile strength, psi (MPa), minimum	6,500 (44.8)	4.6.13 ASTM D638, 20 inches/minute
Ultimate elongation, percent, minimum	200	4.6.13 ASTM D638, 20 inches/minute
Dielectric breakdown volts, minimum	11,000	4.6.2 ASTM D876
Volume resistivity, Ohm-cm, minimum	10^{15}	4.6.2 ASTM D876
Water absorption, percent, maximum	0.1	4.6.2 ASTM D570, 24 hrs. at 23°C
Low temperature flexibility	No cracking	4.6.7.1 -65° ± 1°C (-85° ± 2°F)
Heat resistance, properties after:		4.6.9 200° ± 2°C (392° ± 4°F) for 168 hours
Tensile strength, psi (MPa), minimum	6,000 (41.4)	
Ultimate elongation, percent, minimum	175	
Dielectric breakdown volts, minimum	10,000	
Specific gravity, maximum	1.73	4.6.2 ASTM D792
Flammability	Self-extinguishing 15 seconds and 3 inches	4.6.14 Procedure A ASTM D2671

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Storage life conditions: Supplier shall certify to storage at 18° to 35°C (65° to 95°F) for four years. Conformance to 3.5. See 3.5.2 for storage life extension.

Intended use: Heat shrinkable, semi-rigid, ethylene-tetrafluoroethylene fluoropolymer is intended for use in component cables and electronic lead strain relief where low expansion ratios are satisfactory.

Custodians:
Army - ER
Navy - AS
Air Force - 85

Review activities:
Army - EA, MI
Navy - EC
Air Force - 99
DLA - GS

Preparing activity:
Navy - AS

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